

CUSTOMER AND ORDER MANAGEMENT

Java Programming Assignment

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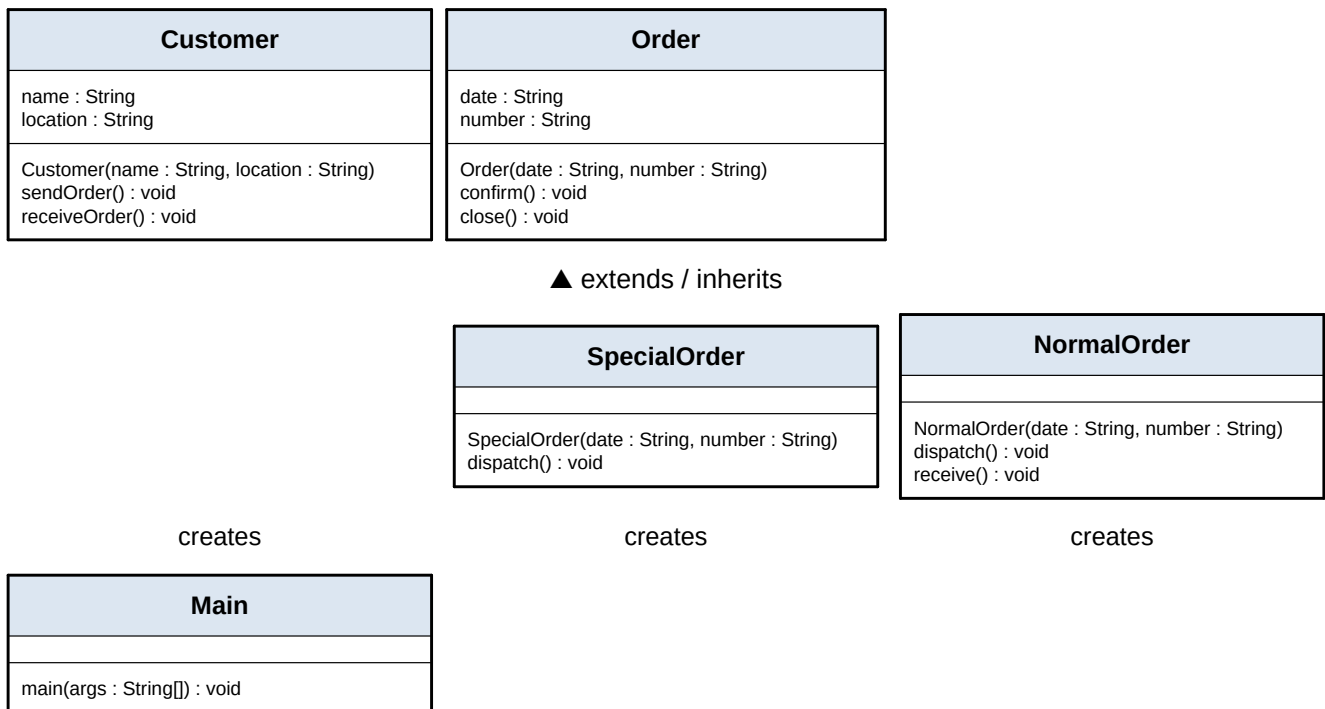
1. ALGORITHM

Steps for executing the Customer and Order Management program:

Step	Algorithm
1	Start the program.
2	Create a Customer object with name "Tharun" and location "Hyderabad".
3	Create a SpecialOrder object with date "16-08-2026" and number "S001".
4	Create a NormalOrder object with date "16-08-2026" and number "N001".
5	Display the customer details.
6	For the special order, send the order, confirm it, dispatch it, close it, and receive it.
7	For the normal order, send the order, confirm it, dispatch it, receive it, close it, and receive it.
8	Display the messages for each operation.
9	Stop the program.

2. UML CLASS DIAGRAM

SpecialOrder and NormalOrder inherit from Order. Main creates Customer, SpecialOrder, and NormalOrder objects.



UML: SpecialOrder and NormalOrder are subclasses of Order. Main uses all three object types.

3. JAVA SOURCE CODE

File name: **Main.java**

```
class Customer {
    String name;
    String location;

    Customer(String name, String location) {
        this.name = name;
        this.location = location;
    }

    void sendOrder() {
        System.out.println(name + " sent an order.");
    }

    void receiveOrder() {
        System.out.println(name + " received the order.");
    }
}

class Order {
    String date;
    String number;

    Order(String date, String number) {
        this.date = date;
        this.number = number;
    }

    void confirm() {
        System.out.println("Order " + number + " is confirmed.");
    }

    void close() {
        System.out.println("Order " + number + " is closed.");
    }
}

class SpecialOrder extends Order {
    SpecialOrder(String date, String number) {
        super(date, number);
    }

    void dispatch() {
        System.out.println("Special order " + number + " is dispatched.");
    }
}

class NormalOrder extends Order {
```

3. JAVA SOURCE CODE (CONTINUED)

```
NormalOrder(String date, String number) {
    super(date, number);
}

void dispatch() {
    System.out.println("Normal order " + number + " is dispatched.");
}

void receive() {
    System.out.println("Normal order " + number + " is received.");
}
}

public class Main {
    public static void main(String[] args) {

        Customer customer = new Customer("Tharun", "Hyderabad");

        SpecialOrder special = new SpecialOrder("16-08-2026", "S001");

        NormalOrder normal = new NormalOrder("16-08-2026", "N001");

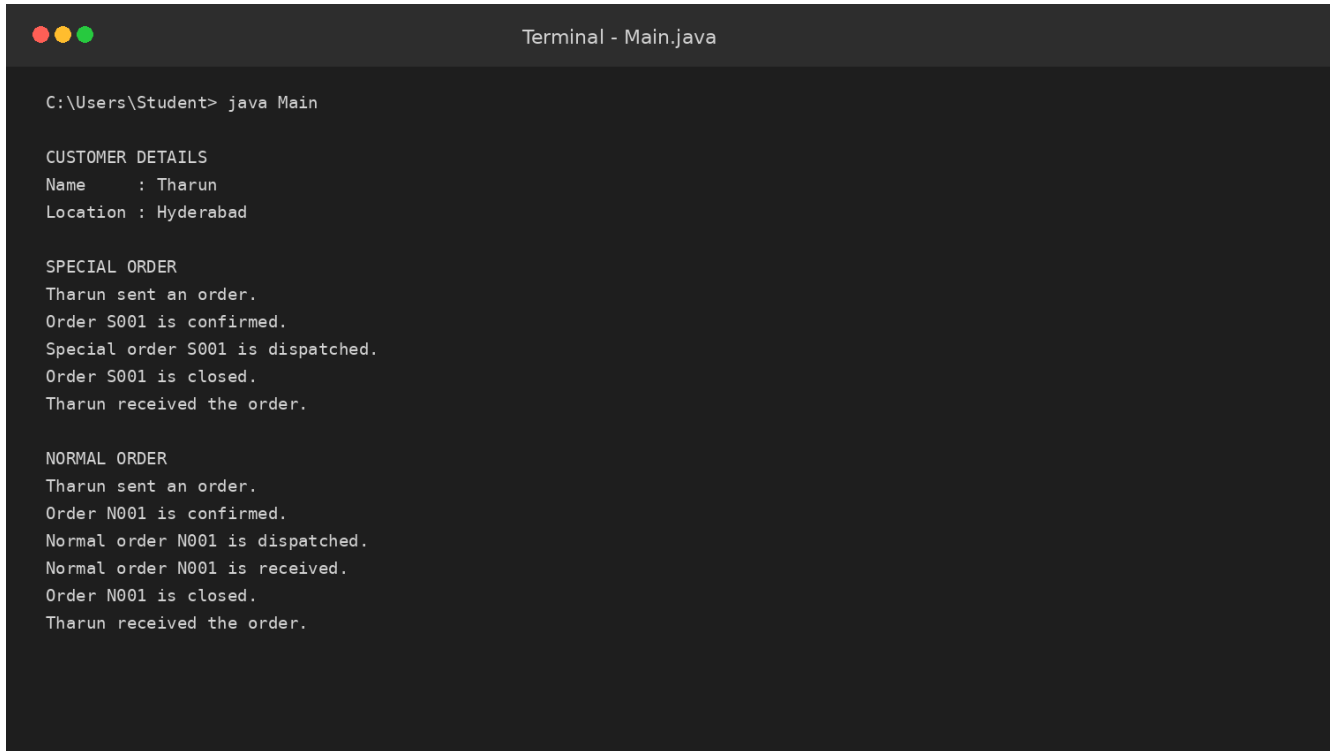
        System.out.println("CUSTOMER DETAILS");
        System.out.println("Name      : " + customer.name);
        System.out.println("Location : " + customer.location);

        System.out.println("\nSPECIAL ORDER");
        customer.sendOrder();
        special.confirm();
        special.dispatch();
        special.close();
        customer.receiveOrder();

        System.out.println("\nNORMAL ORDER");
        customer.sendOrder();
        normal.confirm();
        normal.dispatch();
        normal.receive();
        normal.close();
        customer.receiveOrder();
    }
}
```

4. OUTPUT SCREEN - VS CODE STYLE

Example execution of the program:

A screenshot of a terminal window titled "Terminal - Main.java". The window has a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. The terminal shows the command "C:\Users\Student> java Main" and its output. The output is organized into sections: "CUSTOMER DETAILS" with fields "Name : Tharun" and "Location : Hyderabad"; "SPECIAL ORDER" with a sequence of status messages for order S001; and "NORMAL ORDER" with a sequence of status messages for order N001.

```
C:\Users\Student> java Main

CUSTOMER DETAILS
Name      : Tharun
Location  : Hyderabad

SPECIAL ORDER
Tharun sent an order.
Order S001 is confirmed.
Special order S001 is dispatched.
Order S001 is closed.
Tharun received the order.

NORMAL ORDER
Tharun sent an order.
Order N001 is confirmed.
Normal order N001 is dispatched.
Normal order N001 is received.
Order N001 is closed.
Tharun received the order.
```

Output: Customer details and special/normal order operations are displayed in the terminal.